SAMPLE

The Schrödinger Equation

The Wave Equation of Quantum Mechanics

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Introduction

Every era of physics has produced one equation that came to stand for the whole enterprise. Newton gave mechanics its laws of motion, Maxwell gathered electricity and magnetism into four compact statements, and Einstein bent space and time into a single geometric relation. The twentieth century's contribution came in the winter of 1925, when an Austrian physicist named Erwin Schrödinger, working in Zurich, wrote down a differential equation that describes how the wave function of a physical system changes in time. The Schrödinger equation did for quantum mechanics what Newton's second law had done for classical mechanics: it became the engine of prediction, the statement from which the behavior of matter at its smallest scales could be computed.

The equation's subject is an object that no one has ever seen. The wave function — usually written as a single Greek letter — is not a physical wave in any ordinary sense. It does not carry energy through a medium the way a sound wave carries pressure through air, and it is not a substance that can be collected or weighed. What it carries is possibility. Interpreted according to the rule that Max Born supplied within months of the equation's publication, the wave function's square magnitude tells the probability of finding a particle in one place rather than another. The equation, in turn, tells how that cloud of possibility evolves, smoothly and deterministically, from moment to moment. Quantum mechanics is the only fundamental theory whose central object is a bookkeeping device for what might happen rather than a description of what is there.

This book is an account of that equation: where it came from, what it says, how it was tested, and what it created. The chapters ahead move from the discovery itself — a story that runs through de Broglie's matter waves, a challenge at a Zurich colloquium, and a celebrated skiing holiday — to the equation's content and characteristics, then outward to its consequences: the quantized energy levels of atoms, the chemical bond, the transistor, the tunneling phenomena that power both the stars and the microscope. Later chapters turn to the interpretive puzzles the equation could not settle on its own, the modern research programs that treat it as a computational problem to be solved at scale, and the directions in which its second century of use is now unfolding.

A word about approach. This book describes every mathematical object in words, and no equations appear beyond the equation's own name. The reasoning is that the equation's cultural and scientific significance can be presented entirely in plain language, and that readers who want the

mathematics have, by now, an entire century of textbooks waiting for them. What those textbooks often skip — the historical accidents, the philosophical stakes, the personalities, and the engineering marvels that trace back to a wave equation nobody wanted to love — is precisely the material gathered here. The Schrödinger equation turned a crisis in physics into the most precisely confirmed theory humanity possesses. It also opened questions about the nature of reality that remain open a hundred years later. Both halves of that inheritance are part of the story, and both get their due in the chapters that follow.

Chapter 1: The Discovery of The Schrödinger Equation

The discovery happened at a moment of fertile confusion. By 1924, physics had spent two decades digesting the fact that classical mechanics, so unblemished on the scale of planets and pendulums, failed completely at the scale of atoms. Bohr's 1913 model of the hydrogen atom — an electron circling a proton in orbits of mysteriously permitted sizes — reproduced the hydrogen spectrum but offered no reason for its own rules. Werner Heisenberg, in the summer of 1925, had produced a radical alternative: matrix mechanics, which discarded orbits and pictures altogether and rebuilt mechanics from the observed frequencies and intensities of spectral lines. The matrices worked, but even their creators admitted that the theory was abstract to the point of hostility. It was in this atmosphere, in November 1925, that Erwin Schrödinger read Louis de Broglie's doctoral thesis.

De Broglie's proposal was audacious in the extreme. If light — long considered the purest of waves — behaves in some experiments like a stream of particles, then matter, long considered the purest of particles, should behave in some experiments like a wave. Every moving body, de Broglie argued, has an associated wavelength inversely proportional to its momentum, and for electrons the wavelength is of a size comparable to atomic spacings. The thesis had been received with polite skepticism in Paris; Einstein, consulted by letter, had pronounced it significant, and that judgment carried weight. Schrödinger, then a thirty-eight-year-old professor of theoretical physics at the University of Zurich with a background in wave physics and a lifelong admiration for his countryman Boltzmann's atomic statistics, saw something in de Broglie's pages that others had missed: if the electron is a wave, then the permitted orbits of the Bohr atom might be nothing more mysterious than the standing waves of a plucked string — the only vibrations that fit neatly around a circle.

The Zurich setting supplied the spark. Physics colloquia in Zurich were jointly run by the university and the neighboring ETH, and Peter Debye, Schrödinger's colleague across the street, chaired the seminar. According to the account Debye himself gave decades later, he asked Schrödinger to present de Broglie's work at the seminar, and Schrödinger spent the following weeks searching for the wave equation that de Broglie's thesis implied but did not state. The hunt culminated over the Christmas and New Year holidays of 1925-1926, which Schrödinger spent at a villa in Arosa, a ski resort in the Swiss Alps, accompanied — the biographical record is coy on the point, deliberately so — by a woman whose identity has never been established and

who was not his wife. The episode has become one of physics' most repeated anecdotes, and while the romantic embellishments belong to legend rather than record, the timing is documented: the epochal paper was written in that holiday month.

What Schrödinger produced was a differential equation — the wave mechanics analogue of Newton's law of motion — whose first test was the hydrogen atom. Working it through, he found that the equation yielded, automatically and without Bohr's arbitrary rules, exactly the set of permitted energy levels that the hydrogen spectrum displayed. Where Bohr had postulated allowed orbits, the equation produced quantization as a mathematical consequence, the way a drum skin permits only certain vibration patterns. The discovery electrified him; he later confessed that he had worked the hydrogen problem out in a matter of days during that holiday and had been stunned to find the spectrum falling out. The paper, "Quantization as a Problem of Proper Values," was received by Annalen der Physik on January 27, 1926. Three further parts followed in that single year: the second presented the time-dependent form of the equation, the third unified the approach with matrix mechanics, and the fourth extended the treatment to perturbations and to the Stark effect.

The physics world's reaction split along generational lines. Planck and Einstein, the older generation, received wave mechanics with delight: here at last was a picture, a wave equation one could write on a blackboard and solve like the familiar equations of elasticity and sound. Heisenberg's circle in Göttingen and Copenhagen regarded the new mechanics with suspicion — partly for scientific reasons, since Schrödinger's initial interpretation of the wave as a smeared-out physical electron was soon shown to be untenable, and partly for reasons of pride. Schrödinger himself demonstrated in his third paper that wave mechanics and matrix mechanics were mathematically the same theory in two costumes, a result that ended the rivalry in an odd truce: physicists took the calculational convenience of the wave form and the interpretation of the matrix camp, and the synthesis became what is now simply called quantum mechanics.

The personal bitterness did not subside as quickly as the technical dispute; when the generation met years later, Schrödinger was said to complain that the Göttingen physicists were using his beautiful wave mechanics for their own purposes. He had wanted the electron to be a wave; history decided the wave was something stranger.

The equation's discovery story closes with a Nobel Prize that took seven years to arrive. The 1933 Nobel Prize in Physics was awarded jointly to Schrödinger and to Paul Dirac, the British physicist who had found the equation's relativistic

extension, for "the discovery of new productive forms of atomic theory." The delay from 1926 to 1933 was not neglect but caution: the Nobel committee, conservative by constitution, waited until the theory's predictions had been confirmed to a degree that left no escape. By 1933 they had. The equation had already computed the hydrogen spectrum, the Stark effect, the band structure of molecules, and the shapes of chemical bonds, and experiment agreed with it everywhere it was tested. A century later, the record has not changed: no confirmed deviation from the Schrödinger equation has ever been found in its proper domain, the non-relativistic physics of matter.

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